
◆ IAM SESSION 3 — PREMIUM STUDENT NOTES

Identity & Access Management (AWS Cloud Security)

◆ 1. WHAT IS IAM?

IAM (Identity & Access Management) is an AWS service that controls:

- Who can access AWS resources
- What actions they can perform
- Which services and data they can interact with

Core idea:

IAM is the **identity security layer of the entire AWS cloud platform**.

◆ 2. WHY IAM MATTERS

- Every AWS service depends on IAM
- Misconfigured access = security breach risk
- IAM defines the trust boundary of your cloud

Key principle:

If IAM is compromised, the entire AWS environment is exposed.

◆ 3. OFFICIAL AWS DOCUMENTATION (READ THIS)

Learn directly from AWS:

AWS IAM Official Documentation

◆ 4. IAM CORE COMPONENTS

IAM is built on 3 pillars:

◆ USERS

- Real people or system identities

- Have login credentials
- Used for direct access

◆ ROLES

- Temporary identities
- Used by AWS services (EC2, Lambda, etc.)
- No permanent credentials

◆ POLICIES

- JSON permission documents
- Define what is allowed or denied
- Attached to users, roles, or groups

◆ 5. IAM USERS

- Represents a real identity
- Has username, password, or access keys
- Used for signing into AWS

Risk:

Long-term credentials can be exposed or leaked.

Best practice:

Avoid using users for applications in production.

◆ 6. IAM ROLES

- Temporary access mechanism
- No static credentials
- Assumed when needed

Examples:

- EC2 accessing S3
- Lambda accessing databases
- Cross-account access

Key idea:

Roles are **more secure than users**.

◆ 7. IAM POLICIES

Policies define permissions in AWS.

They answer:

- What can this identity do?
- Which resource can it access?
- Is it allowed or denied?

Structure:

- Effect → Allow / Deny
- Action → API permission (e.g. S3:GetObject)
- Resource → Target service or object

◆ 8. LEAST PRIVILEGE PRINCIPLE

Definition:

Give only the permissions required — nothing more.

Why it matters:

- Reduces attack surface
- Limits damage from compromised accounts
- Improves compliance and governance

Bad practice:

Full administrator access for all users

Good practice:

Minimal, role-based access only

◆ 9. MFA (MULTI-FACTOR AUTHENTICATION)

Adds an extra layer of identity protection.

Login requires:

- Password (something you know)
- Device verification (something you have)

Benefit:

Prevents unauthorized access even if password is stolen.

◆ 10. ROOT ACCOUNT RISKS

The AWS root account has unrestricted access.

It can:

- Delete all resources
- Change billing
- Override all IAM policies

Best practice:

- Never use root for daily work
- Enable MFA immediately
- Store credentials securely

◆ 11. COMMON IAM MISTAKES

Real-world security failures:

- Over-permissioned roles
- Shared user accounts
- Missing MFA
- Exposed access keys

Result:

Cloud breaches, data leaks, financial loss

◆ 12. AWS TRAINING VIDEO (WATCH THIS)

[AWS IAM Introduction (Free Training Video)]

◆ 13. IAM IN SIMPLE TERMS

IAM is the cloud's security control system:

- Identity = Who you are
- Policy = What you can do
- Role = Temporary access
- MFA = Extra verification

◆ 14. KEY TAKEAWAY

IAM is the foundation of AWS security.

- Strong IAM → Secure cloud
- Weak IAM → Full exposure risk
